

Math 526, S05
Quiz 5

Name:
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (14 points) Let $\mathbf{x} = \begin{bmatrix} 1 \\ -2 \\ 2 \\ 7 \end{bmatrix}$, $\mathbf{A} = \begin{bmatrix} -1 & 2 & 3 & 1 \\ 2 & 3 & -4 & 1 \\ 2 & -5 & -2 & 1 \\ -1 & -1 & -1 & 2 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$.

a) Compute $\|\mathbf{x}\|_1$, $\|\mathbf{x}\|_2$ and $\|\mathbf{x}\|_\infty$.

Solution:

$$\|\mathbf{x}\|_1 = 1 + 2 + 2 + 7 = 12.$$

$$\|\mathbf{x}\|_2 = \sqrt{1^2 + (-2)^2 + 2^2 + 7^2} = \sqrt{58}.$$

$$\|\mathbf{x}\|_\infty = \max\{1, 2, 2, 7\} = 7.$$

b) Compute $\|\mathbf{A}\|_1$ and $\|\mathbf{B}\|_2$.

Solution:

$$\|\mathbf{A}\|_1 = \max\{1 + 2 + 2 + 1, 2 + 3 + 5 + 1, 3 + 4 + 2 + 1, 1 + 1 + 1 + 2\} = 11.$$

$$\|\mathbf{B}\|_2 = \max\{1, 3, 5\} = 5.$$

Problem 2. (6 points) Describe in details the three basic properties of matrix norm: *positivity*, *homogeneity* and *triangle inequality*.

Solution:

Positivity: $\|\mathbf{A}\| \geq 0$ and $\|\mathbf{A}\| = 0$ if and only if $\mathbf{A} = \mathbf{0}$.

Homogeneity: $\|\alpha\mathbf{A}\| = |\alpha|\|\mathbf{A}\|$ for any scalar α .

Triangle inequality: $\|\mathbf{A} + \mathbf{B}\| \leq \|\mathbf{A}\| + \|\mathbf{B}\|$.